

Security Operations Center - Threat Analysis Report

Threat Level: LOW | **Total Alerts:** 2 **Generated:**
2025-11-16T09:04:56.091Z

Executive Summary

Two medium-severity Suricata alerts were detected involving TCPv4 invalid checksums from a single external source (66.96.202.66) in Singapore. Both alerts originated from the same source IP, targeting distinct external destinations in the United States over TLS (port 443). The alerts were flagged as external threats with low confidence, indicating potential network-level anomalies or malformed traffic. No internal infrastructure or lateral movement was observed. The traffic patterns suggest possible scanning or protocol manipulation attempts, though no malicious payload or behavioral indicators were detected. No immediate exploitation or data exfiltration is evident. Further investigation into the source IP and its historical activity is recommended.

Key Findings

- Two identical Suricata alerts (TCPv4 invalid checksum) from 66.96.202.66 (Singapore) targeting US-based external endpoints.
- Both alerts occurred within 9 seconds, indicating a coordinated or rapid sequence of network probes.
- Traffic was encrypted (TLS), limiting deep packet inspection capabilities.
- No evidence of malware C2, data exfiltration, or lateral movement detected.
- Source IP not previously associated with known malicious activity in available context.

Top Priority Threats

IP Address	Type	Country	Direction	Activity	Severity	Confidence	Count
66.96.202.66	External	Singapore	Inbound	TCP Checksum Anomaly	CRITICAL	High	1

MITRE ATT&CK Mapping

- **T1048** - Exfiltration Over Alternative Protocol (Exfiltration)
- **T1071.004** - DNS (Command And Control)
- **T1595** - Active Scanning (Reconnaissance)

Immediate Actions Required

1. Block outbound traffic from 66.96.202.66 at the firewall if not already in place.
2. Monitor for additional connections from this IP to internal or external services.
3. Cross-reference 66.96.202.66 against threat intelligence feeds (e.g., AlienVault OTX, VirusTotal).
4. Review TLS session logs for abnormal handshake patterns or certificate anomalies.
5. Document and archive alert data for future correlation with similar patterns.

Analysis Complete Report generated: 2025-11-16 17:04:59 Threat level: LOW Priority actions: 0 identified